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58. Let $A = A_{180^\circ}$, and let B be the matrix that reflects R^2 about the x-axis; that is,

$$B = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \qquad A = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

Compute BA , and describe geometrically how a vector v is affected by multiplication by BA .

60. Let A be an $n \times n$ matrix. If A and B are $n \times n$ diagonal matrices, then AB is a diagonal matrix whose j th column is $a_{jj}b_{jj}\mathbf{e}_j$.

59. A square matrix A is called **lower triangular** if the (i, j) -entry of A is zero whenever $i < j$. Prove that if A and B are both $n \times n$ lower triangular matrices, then AB is also a lower triangular matrix.

61. A square matrix A is called **upper triangular** if the (i, j) -entry of A is zero whenever $i > j$. Prove that if A and B are both $n \times n$ upper triangular matrices, then AB is also an upper triangular matrix.

62. Let $A = \begin{bmatrix} 1 & -1 & 2 & -1 \\ -2 & 1 & -1 & 3 \\ -1 & -1 & 4 & 3 \\ -5 & 3 & -4 & 7 \end{bmatrix}$.

Find a nonzero 4×2 matrix B with rank 2 such that $AB = O$

63. Find an example of $n \times n$ matrices A and B such that $AB = 0$, but $BA \neq 0$.

$$A = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$AB = 0$$

$$BA = \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix}$$

$$\begin{matrix} \checkmark & \searrow \\ \begin{bmatrix} 0 & 0 \\ 1 & 0 \end{bmatrix} & \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix} \end{matrix}$$

64. Let A and B be $n \times n$ matrices. Prove and disprove that the ranks of AB and BA are equal.

$$\begin{matrix} \Rightarrow 0 & \neq 0 \end{matrix}$$

$$\begin{matrix} AB = 0 \\ BA \neq 0 \end{matrix}$$

$$\begin{matrix} \text{rank } AB = 0 \\ \text{rank } BA > 0 \end{matrix}$$

65. Recall the definition of trace of a matrix. Prove that if A is an $m \times n$ matrix and B is an $n \times m$ matrix, then $\text{trace}(AB) = \text{trace}(BA)$.

$$A: n \times n$$

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \dots & a_{nn} \end{bmatrix}$$

$$n \times n$$

$$\text{trace } A = a_{11} + a_{22} + \dots + a_{nn}$$

$$AB \neq BA$$

$$\text{trace}(AB)$$

$$= \text{trace}(BA)$$

66. Let $1 \leq r, s \leq n$ be integers, and let E be the $n \times n$ matrix with 1 as the (r, s) -entry and 0s elsewhere. Let B be any $n \times n$ matrix. Describe EB in terms of the entries of B .

68. (a) Let A and B be symmetric matrices of the same size. Prove that AB is symmetric if and only if $AB = BA$.
- (b) Find symmetric 2×2 matrices A and B such that $AB \neq BA$.